

DESIGN PORTFOLIO



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Industrial and Production Engineer



Certified SOLIDWORKS CAD DESIGN Associate



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Ergonomic Medical Chair (ENT/Dental)

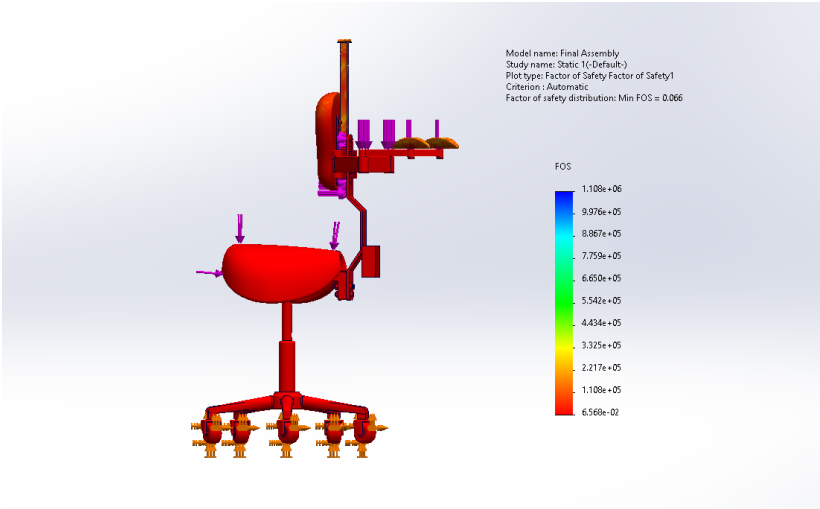
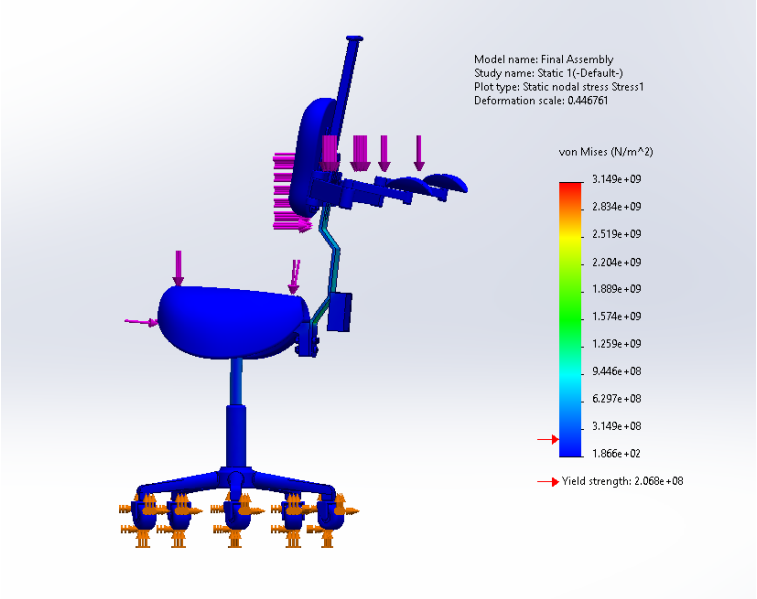
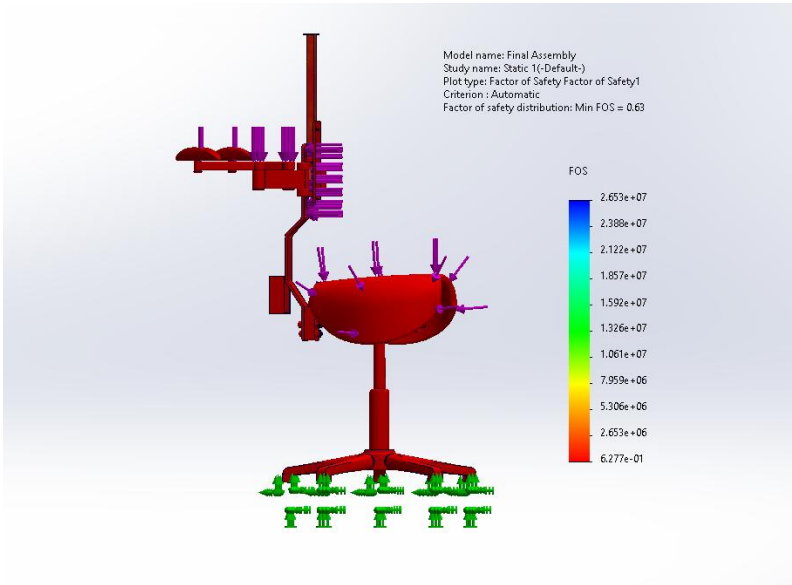
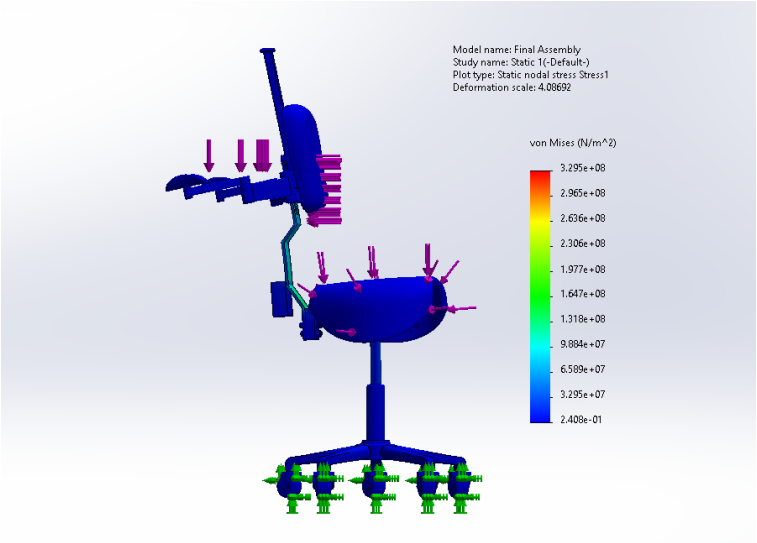
Designed and developed an ergonomic medical chair for **ENT and dental applications**, focusing on patient comfort, practitioner accessibility, and structural stability. The design incorporates **human-centered ergonomic principles** to improve adjustability, usability, and overall user experience.

Key Features & Design Considerations:

- Developed a comfortable and ergonomically optimized seating structure for medical procedures.
- Focused on **patient posture, accessibility, and operator convenience**.
- Incorporated adjustable elements to accommodate different users and treatment requirements.
- Applied **human-centered design and ergonomic principles** throughout the product development process.
- Considered structural stability, functionality, and manufacturability in the design.
- The project was recognized with **2nd Place at a Product Showcase** for its innovative and practical design approach.



FEA Analysis

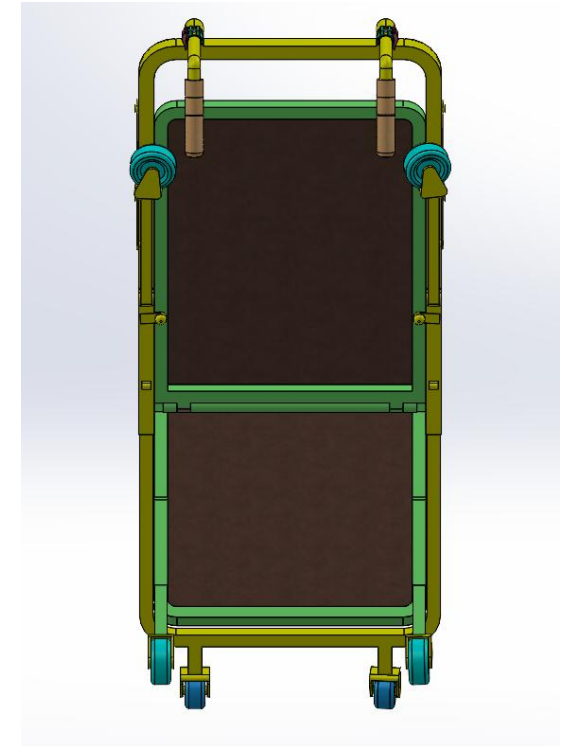
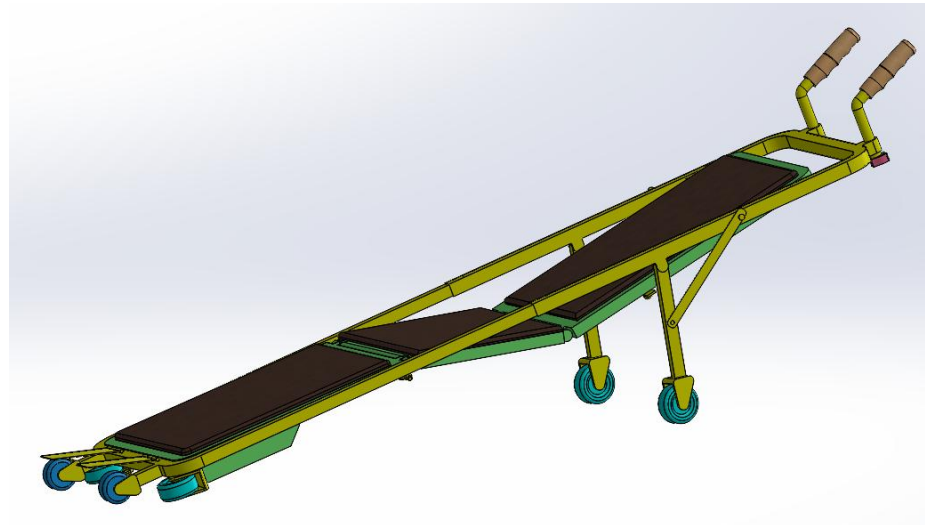


Ergonomic Stretcher-cum-Wheelchair

Designed and developed a multifunctional **stretcher-cum-wheelchair** to improve patient transportation and mobility in healthcare environments. The concept combines the functionality of a stretcher and wheelchair into a single ergonomic system, enabling easier patient handling and movement.

Key Features & Design Considerations:

- Developed a convertible design integrating **stretcher and wheelchair functionality**.
- Focused on **patient safety, comfort, mobility, and ease of operation**.
- Applied ergonomic principles to improve handling for both patients and healthcare practitioners.
- Considered structural stability and practical usability during transportation.
- Designed the system with a focus on **efficient patient transfer and maneuverability**.
- Recognized with **3rd Place at a Product Showcase** for its practical and user-centered design approach.

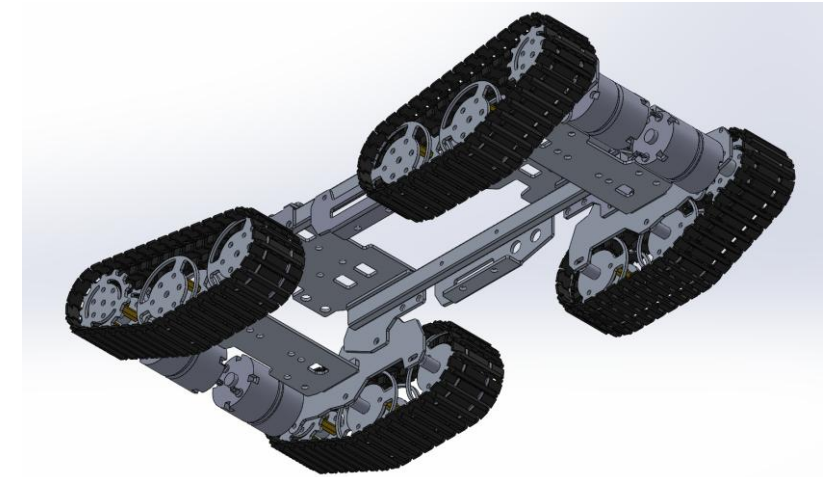
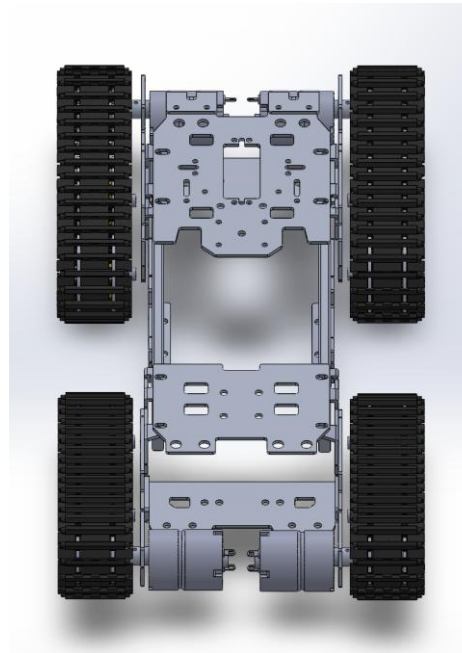
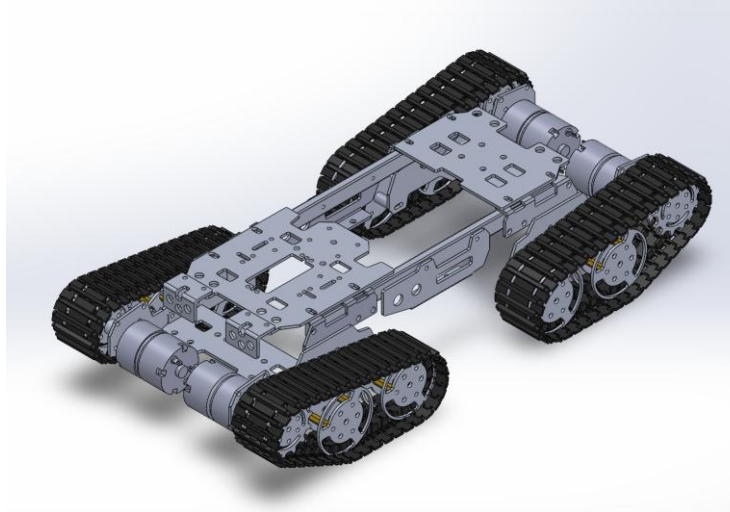


4-Track Robotic UGV Chassis

Designed and developed a **modular four-track Unmanned Ground Vehicle (UGV) chassis** to improve mobility, stability, and terrain adaptability for operation on rough and uneven terrain. The design focused on creating a robust and maintainable platform capable of handling challenging ground conditions.

Key Features & Design Considerations:

- Developed a **four-track modular chassis** for enhanced traction, stability, and off-road mobility.
- Focused on **structural robustness and load-bearing capability** for reliable operation on rough terrain.
- Incorporated a **modular design approach** to simplify assembly, maintenance, and future modifications.
- Considered **terrain adaptability and ground clearance** to improve vehicle performance over uneven surfaces.
- Designed with emphasis on **stability and maneuverability** during operation on challenging terrain.
- Prioritized **maintainability and component accessibility** for efficient servicing and replacement.

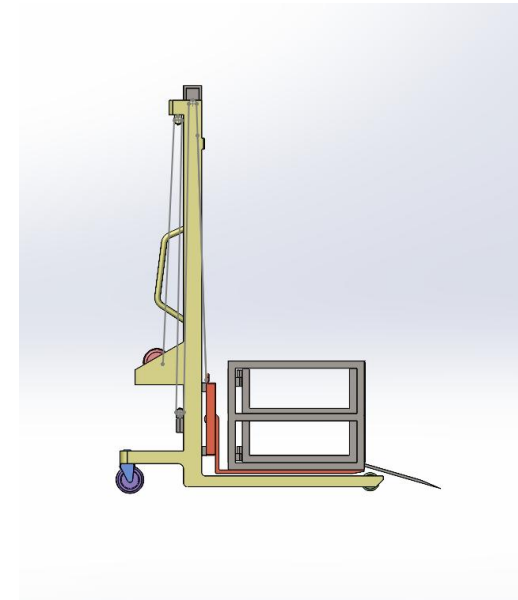
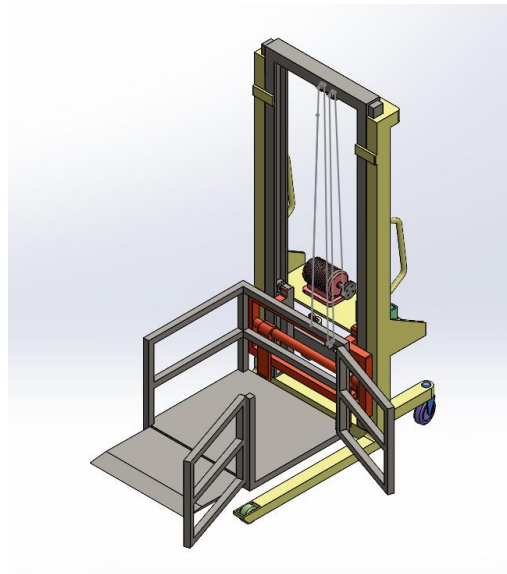
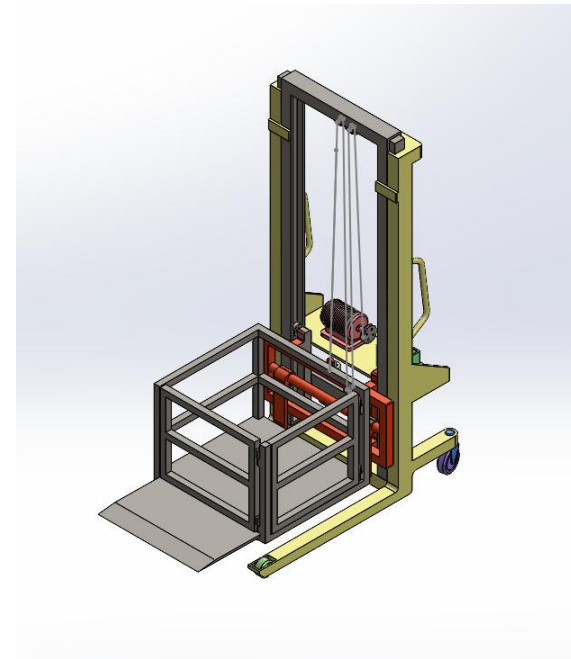
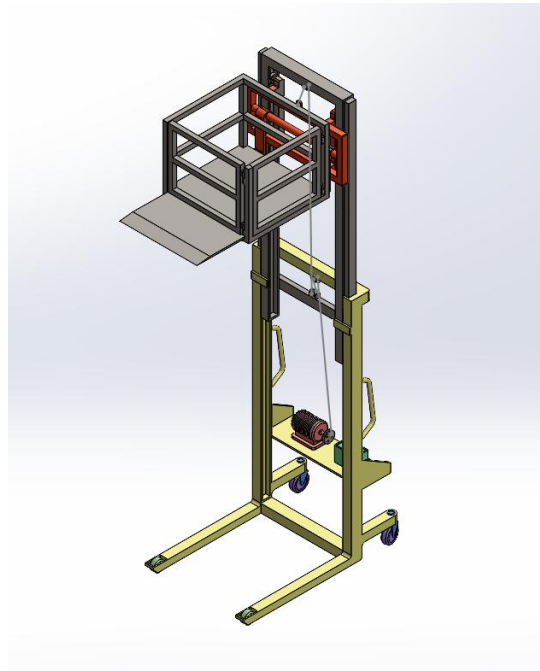


Hydraulic Double Chamber Power Lifter

Designed and developed a **hydraulic material lifting system** incorporating a double-chamber lifting mechanism to improve load-carrying capacity, lifting performance, and operational reliability.

Key Features & Design Considerations:

- Developed a **double-chamber hydraulic lifting mechanism** for enhanced load-bearing capability.
- Focused on **hydraulic power transmission and structural design** to ensure efficient lifting operation.
- Considered **structural strength and stability** to withstand heavy lifting loads.
- Designed with emphasis on **reliable and controlled lifting performance** during operation.
- Incorporated hydraulic system principles to improve **operational efficiency and lifting capacity**.
- Prioritized **robustness, safety, and practical applicability** for material-handling applications.



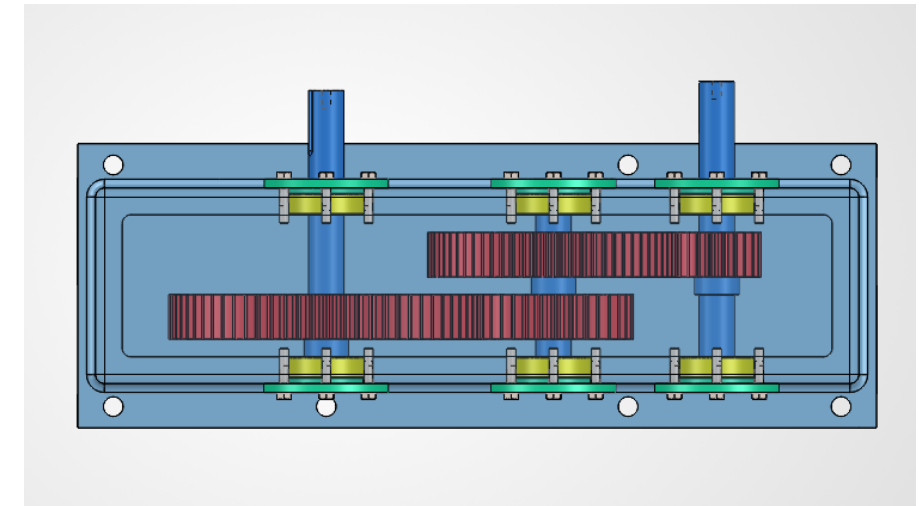
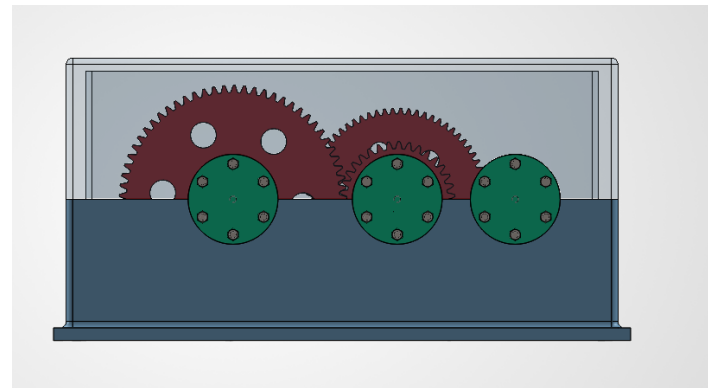
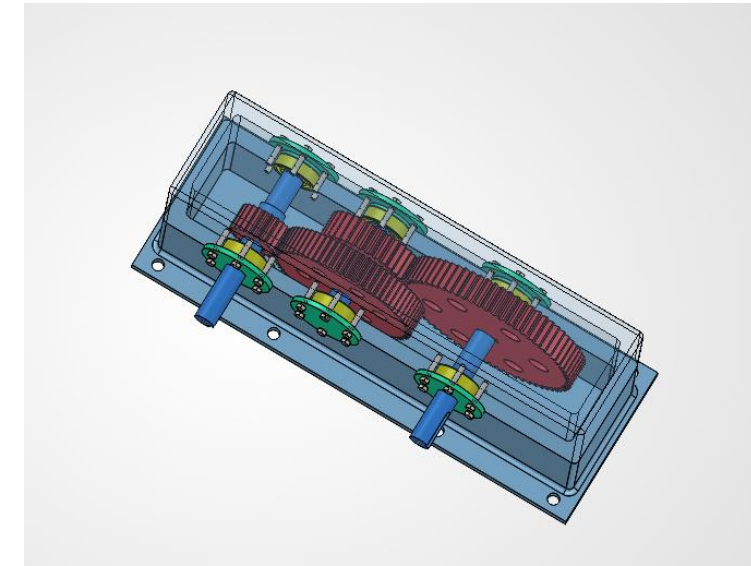
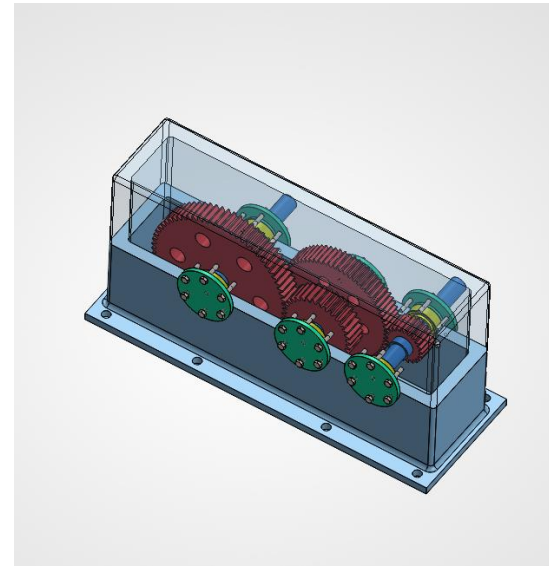
Compact Three-Stage Spur Gear Transmission System

Project #05

Designed and developed a **compact enclosed three-stage spur gear transmission system** for efficient speed reduction and torque multiplication, with emphasis on mechanical performance, durability, and manufacturability.

Key Features & Design Considerations:

- Developed a **three-stage spur gear transmission** to achieve effective speed reduction and increased output torque.
- Designed **gear assemblies, shafts, housing, and supporting components** for reliable power transmission.
- Focused on **compact and enclosed construction** to optimize space utilization and protect internal components.
- Considered **mechanical strength, alignment, and load distribution** for smooth and reliable operation.
- Prioritized **durability and manufacturability** for practical mechanical applications.
- Emphasized **efficient power transmission and structural integrity** throughout the system.



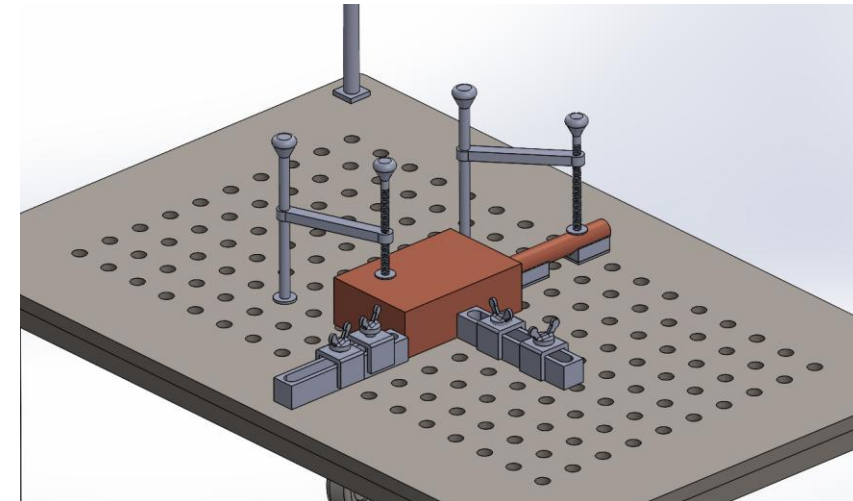
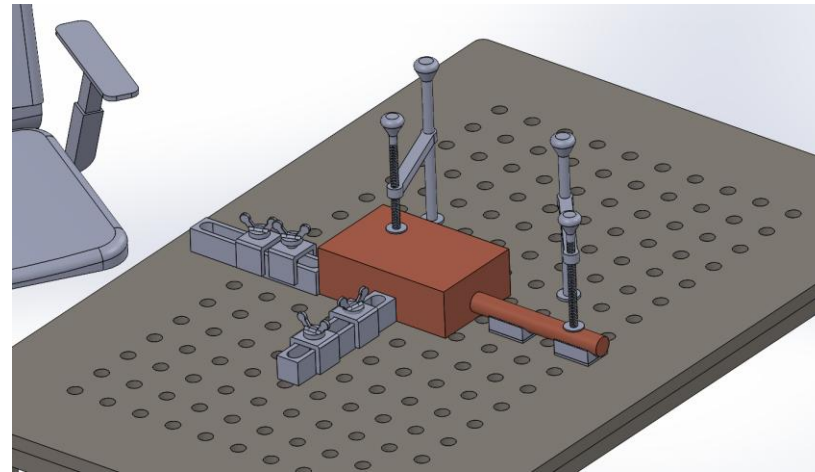
Modular Welding Station

Project #06

Designed and developed a **modular welding workstation** with integrated tool organization and ergonomic accessibility to improve workflow efficiency, operator safety, and overall productivity.

Key Features & Design Considerations:

- Developed a **modular workstation layout** to support flexible welding operations and workspace organization.
- Integrated **tool storage and organization features** for easy access and improved workplace efficiency.
- Focused on **ergonomic accessibility** to reduce unnecessary movement and improve operator comfort.
- Optimized the workstation layout to enhance **workflow and production efficiency**.
- Considered **operator safety and workspace organization** for a more effective working environment.
- Prioritized **modularity, accessibility, and practical usability** for industrial welding applications.



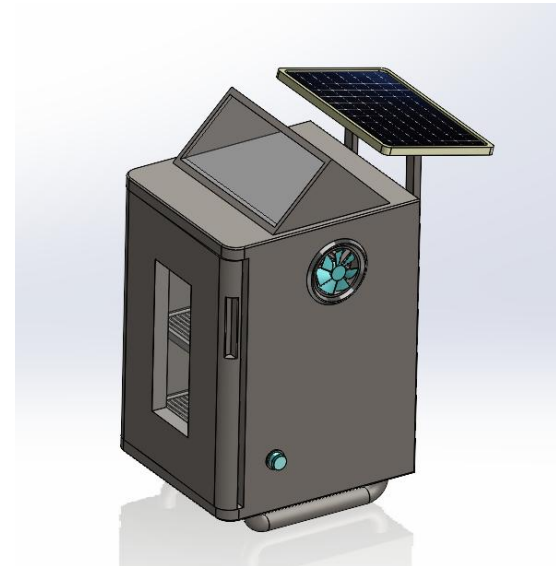
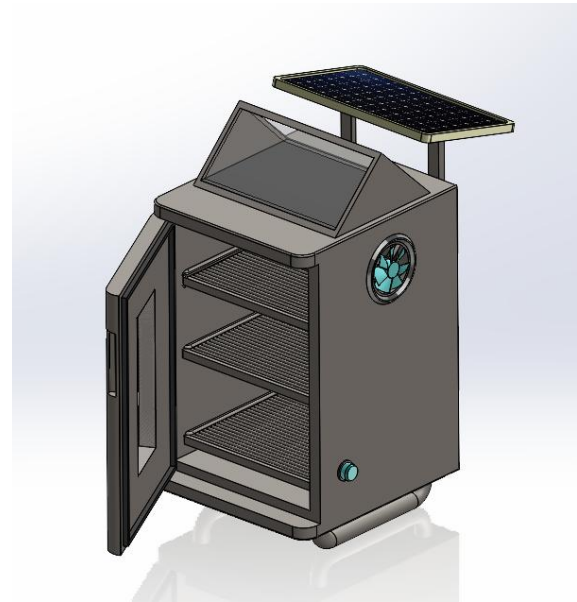
Solar-Integrated Utility Enclosure

Project #07

Designed and developed a **solar-powered smart utility enclosure** for autonomous operation in off-grid environments, integrating photovoltaic energy concepts with a functional industrial enclosure for sustainable and reliable power generation.

Key Features & Design Considerations:

- Developed a **solar-integrated enclosure system** to support autonomous operation in remote and off-grid locations.
- Integrated **photovoltaic system concepts** for sustainable and renewable energy utilization.
- Focused on **compact and functional enclosure design** for efficient integration of electrical and utility components.
- Considered **structural protection and environmental adaptability** for reliable outdoor operation.
- Prioritized **energy sustainability and operational independence** through solar power integration.
- Emphasized **practical usability, durability, and efficient space utilization** for industrial applications.



Project #07

von Mises (N/m²)

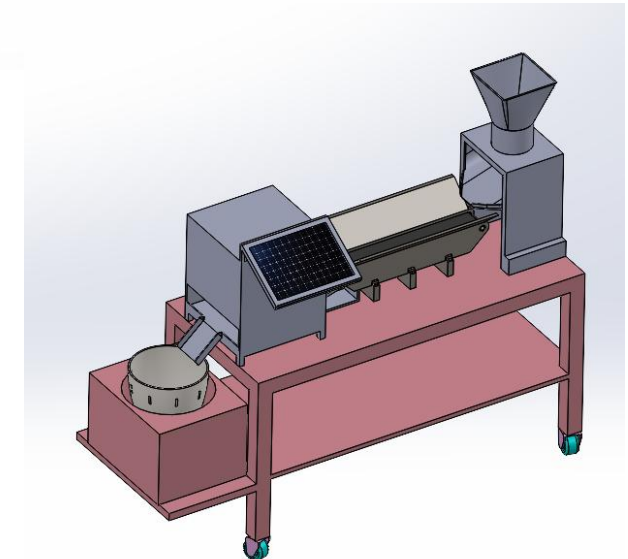
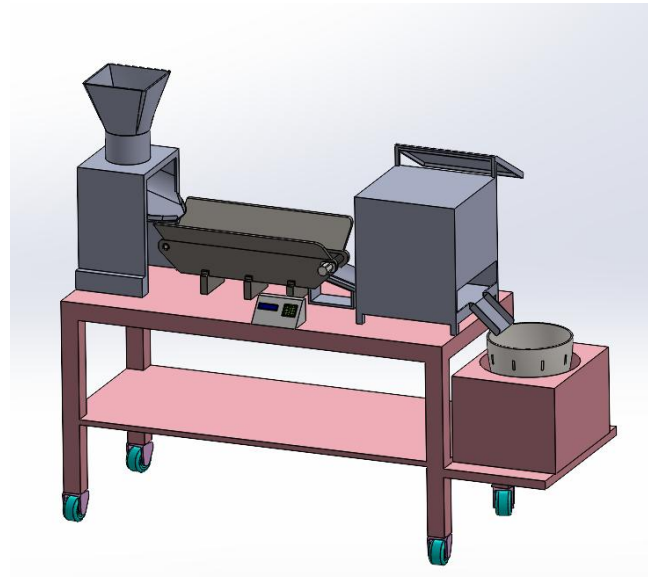
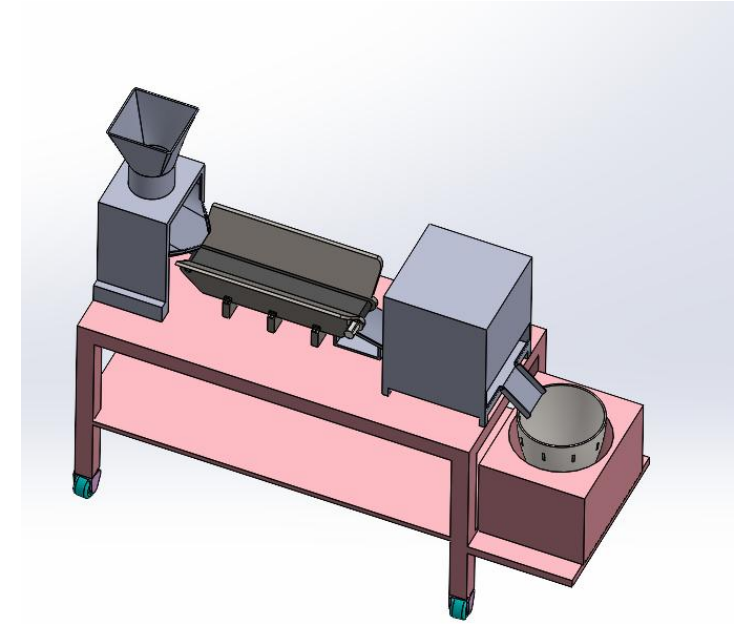
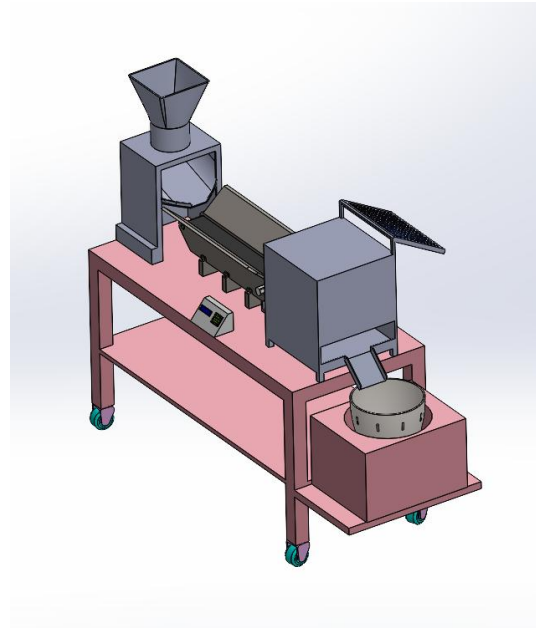
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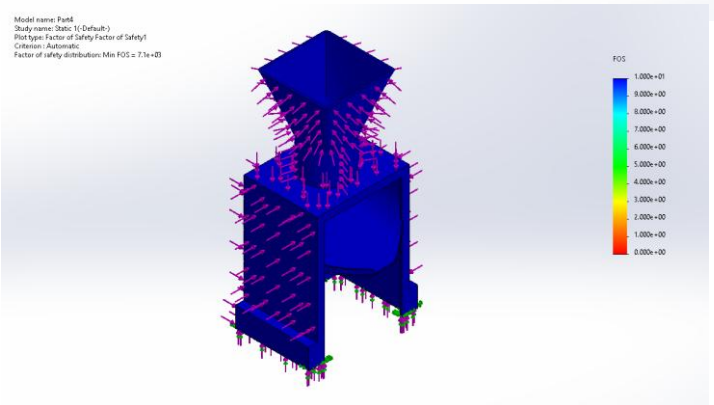
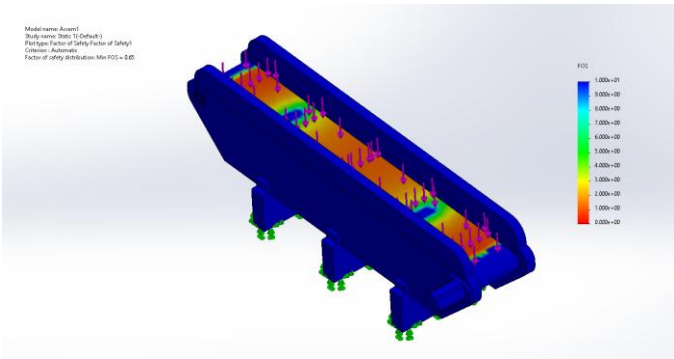
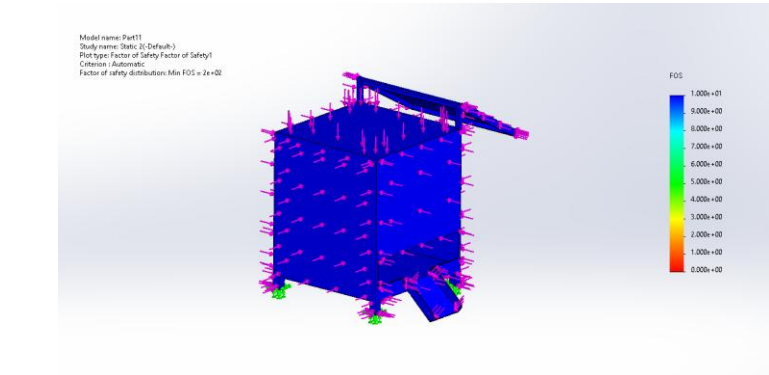
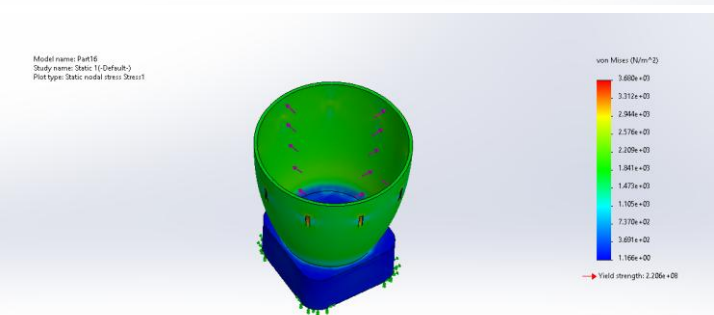
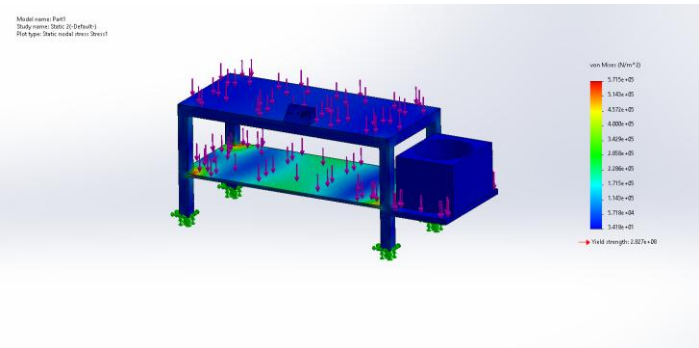
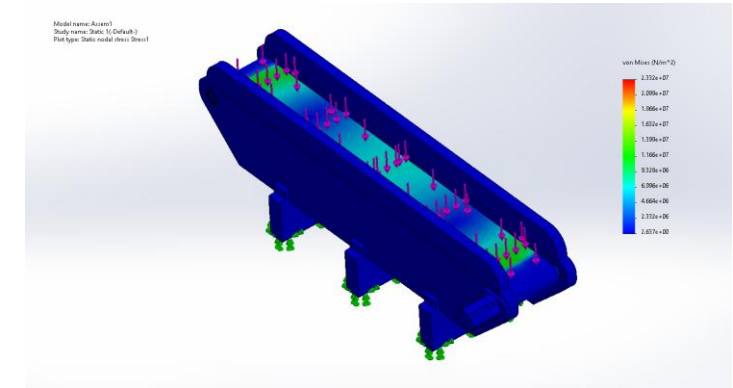
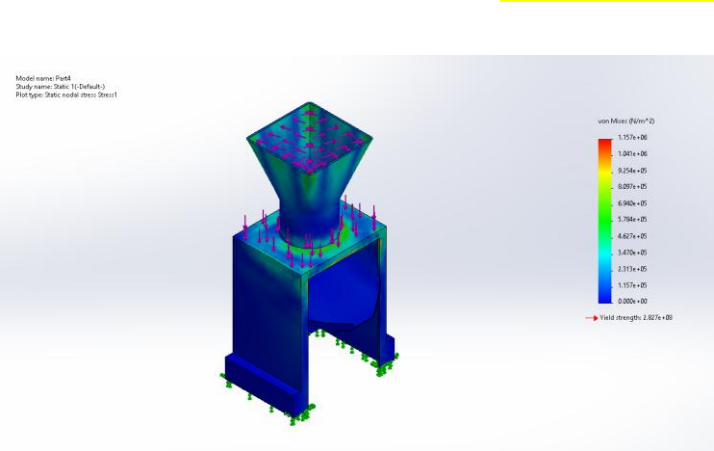
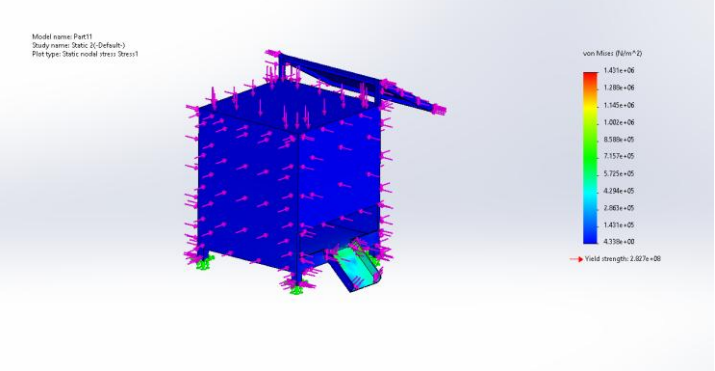
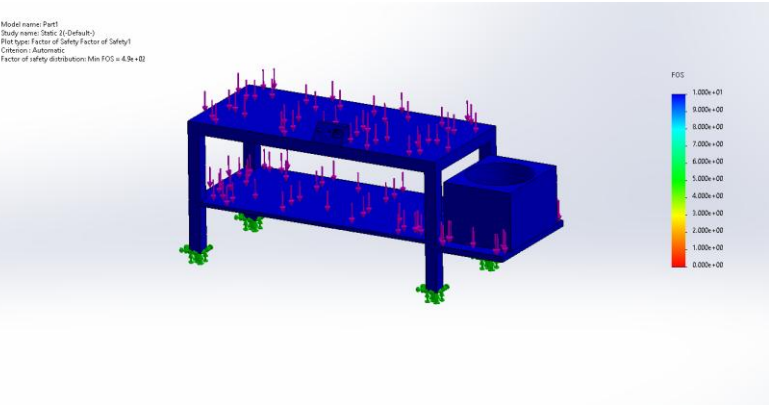
Automatic Eco-Crop Grind Machine

Designed and developed an **automated agricultural grinding machine** to improve processing efficiency, reduce manual effort, and enable reliable continuous operation for agricultural material processing.

Key Features & Design Considerations:

- Developed an **automated grinding mechanism** to improve processing efficiency and minimize manual intervention.
- Designed the **mechanical structure and functional layout** for smooth and reliable operation.
- Focused on **continuous material processing** to enhance productivity and operational efficiency.
- Considered **structural stability and component integration** for dependable machine performance.
- Optimized the system for **ease of operation and practical agricultural applications**.
- Prioritized **automation, durability, and efficient material processing** for improved overall productivity.





Additional Design & Development Projects

